# The discovery of the cell



The first cell ever observed and illustrated was the empty, dead shell of a once living plant. In 1665 the English scientist Robert Hooke (1635–1703) used a reasonably new invention, the microscope, to look closely at a thinly sliced piece of cork. The empty 'shell' of what was once living plant matter brought to mind tiny rooms called cells (from the Latin, *cellula*, meaning 'small compartment'). Some books state that Hooke named them after the cells of monasteries, while others cite the cells of honeycombs, but whatever inspired him, Hooke named them well.

It took another 170 years before the living contents of cells were discovered. Bohemian physiologist Johannes von Purkinje (1787–1869) recognised that cells had to be composed of a living matter since they formed a part of living matter.* He named the matter *protoplasm* (*proto* first, *plasm* form). French scientist Felix Dujardin (1801–1860) eventually observed the protoplasm. Dujardin called it *sarcode*, from the Greek *sarkos*, meaning 'flesh'.

** Matthias Schleiden (1804–1881) confirmed this for plants in 1838.*

Not long after Dujardin observed the protoplasm, German scientist Theodor Schwann (1810–1882) made the first really detailed inspection of animal cells. It soon became clear that both plants and animals possess cells, each with structures that were unique to the group and many that were common to both. It was Schwann who showed us that every tissue of every organism is made of cells. Prior to this many people recognised that cells formed a *part* of an organism, but they did not realise that every structure could be reduced to cellular components. It was then understood that the husk-like walls, which formed the once living cells of the tree, had created the cork.

The scientists named above provide an excellent example of how the different branches of science can unite to advance our knowledge of a given topic. Give a brief description for the different branches of science each man specialised in.

1 Robert Hooke – Experimental physicist _____

2 Johannes von Purkinje – Physiologist _____

3 Theodor Schwann – Zoologist _____

4 Matthias Schleiden – Botanist _____

5 Felix Dujardin – Biologist _____

6 Write brief summaries about how the works of these men led to the discovery and understanding of the cell.

**Research**

7 Schwann and Purkinje had the following named in their honour: *Schwann cells*, *Purkinje cells* and *Purkinje fibres*. Investigate and briefly describe each.

## The Cell Theory

When it became clear that all living things were made of cells, scientists and naturalists began taking a close look at all kinds of other things. Thanks to the microscope, tiny worlds, invisible to the naked eye, were finally revealed. One such world was the minute life forms found in pond water. Some were made of a single cell and were called *unicellular* organisms (*uni* one) while those made of many cells were named *multicellular* organisms (*multi* many).

The statement that all living things are made of cells was known as *The Cell Theory*. The original theory has been modified to state that the cell is the basic primary structure of all living things. This statement recognises that not all components of living organisms contain living cells. For example, black bread moulds and certain by-products of cellular activity do not. The newer theory also states that living cells can only arise from other living cells and that the cell is the smallest form of life.

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We now accept that all living (organic) matter, from the tiny unicellular organisms to the largest multicellular organisms, is made up of basically the same components. These 'core' components are non-living (inorganic) matter. It is the arrangement of this matter that differs between organisms.

In 1855, Rudolf Virchow (1821–1902) the German anthropologist, pathologist, physician and physiologist, added to the cell theory. He came up with the statement '*Omnis cellula e cellula*', or 'all cells develop only from existing cells'. The smallest cells are those of bacteria and they range from just over 1 to about 10 micrometres (1 micrometre = $10^{-6}$ m = 1 millionth of a metre). The largest cells are the eggs of large birds and reptiles, which can be up nearly 10 cm in diameter.

**8** Summarise *The Cell Theory* in point form.

**9** *The cell is the smallest form of life.* Discuss this statement.

**10** Give one example each of a plant and an animal by-product that can be described as an exception to the cell theory.

**11** Give one example each of *a unicellular* organism and *a multicellular* organism.

**12** Suggest why unicellular organisms cannot grow to be as large as multicellular organisms.

**13 Mental challenge!** Explain how living (organic) matter can arise from non-living (inorganic) matter.

**14** Biologists often organise their study of life (and components of living things) in levels of complexity or hierarchies. Arrange the following from simplest to most complex:

*cell* *atom* *organ* *tissue* *organ system* *multicellular organism*

Explain the difference between:

**15** organ and organelle

**16** chlorophyll and chloroplast

**17** molecule and compound

Answer true or false to statements (18–26) below.

**18** Felix Dujardin was the first scientist to see *protoplasm*. _____

**19** Cork was the first substance to reveal cells. _____

**20** All components of living organisms contain living cells. _____

**21** The first cell was observed in 1869. _____

**22** Theodor Schwann made the first detailed observation of animal cells. _____

**23** Rudolph Virchow recognised that cells arise from living cells. _____

**24** A crocodile egg is a cell. _____

**25** Bacteria are the smallest form of life. _____

**26** The first cell observed was a living plant cell. _____

**27** Suggest ways in which the scientists described in questions 1–5 might have contributed to our knowledge of the cell.



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